

3 An infinite horizon problem

Why should macroeconomists study the case of an infinite time horizon?

1. Evolution of the capital is distorted at the end of the model in finite version.
2. Simplicity: dynamic programming is easier.

$$\sum_{t=0}^{\infty} \beta^t u(c_t),$$

where $\beta \in (0, 1)$ is called the discount factor.

Is the infinite sum always convergent?

Preference requirements Consider a plan specifying a constant flow of consumption:

$$\sum_{t=0}^{\infty} \beta^t u(c_t) = \sum_{t=0}^{\infty} \beta^t u(\bar{c}),$$

which is convergent if and only if $\beta < 1$.

Alternatively, consider a constantly increasing consumption stream:

$$c_t = c_0(1 + \gamma)^t.$$

In the case of CES utility representation, boundedness requires $\beta(1 + \gamma)^{1-\sigma} < 1$.

THEOREM 3 (SUFFICIENT CONDITIONS) *If an allocation in the social planner's problem satisfies the Euler equation and a transversality condition*

$$\lim_{t \rightarrow \infty} \beta^t u'(c_t^*) k_{t+1}^* = 0,$$

then it is optimal.

PROOF OF : Stokey, Lucas and Prescott, Theorem 4.15, page 98.

1. $u(F(k_t, 1) + (1 - \delta)k_t - k_{t+1})$ is concave in (k_t, k_{t+1}) .

2.

$$\begin{aligned}
u(F(k_0, 1) + (1 - \delta)k_0 - k_1) &\leq u(c_0^*) - u'(c_0^*)(k_1 - k_1^*) \\
\beta u(F(k_1, 1) + (1 - \delta)k_1 - k_2) &\leq \beta u(c_1^*) + \beta u'(c_1^*)[F_k(k_1^*, 1) + (1 - \delta)](k_1 - k_1^*) \\
&\quad - \beta u'(c_1^*)(k_2 - k_2^*) \\
\beta^T u(F(k_T, 1) + (1 - \delta)k_T - k_{T+1}) &\leq \beta^T u(c_T^*) + \beta^T u'(c_T^*)[F_k(k_T^*, 1) + (1 - \delta)] \\
&\quad - \beta^T u'(c_T^*)(k_{T+1} - k_{T+1}^*)
\end{aligned}$$

3.

$$\begin{aligned}
\sum_{t=0}^T \beta^t u(c_t) &\leq \sum_{t=0}^T \beta^t u(c_t^*) - \beta^T u'(c_T^*)(k_{T+1} - k_{T+1}^*) \\
&\leq \sum_{t=0}^T \beta^t u(c_t^*) + \beta^T u'(c_T^*)k_{T+1}^*
\end{aligned}$$

Taking limit $T \rightarrow \infty$ yields the result.

Q.E.D.

EXAMPLE 4 *Find out*

$$\begin{aligned}
V^*(k_0) &= \sup_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_t \beta^t \log(Ak_t^\alpha - k_{t+1}) \\
s.t. \quad &k_{t+1} \in [0, Ak_t^\alpha] \\
&k_0 \text{ is given.}
\end{aligned}$$

Look at the policy $k_{t+1} = \alpha\beta Ak_t^\alpha$. verify

1. *euler equation*

$$\frac{1}{(1 - \alpha\beta)Ak_t^\alpha} = \frac{\beta\alpha Ak_{t+1}^{\alpha-1}}{(1 - \alpha\beta)Ak_{t+1}^\alpha}$$

2. *transversality condition*. $\lim_t k_t \rightarrow k^* = (A\alpha\beta)^{\frac{1}{1-\alpha}}$, $\lim_t c_t \rightarrow c^* = (1 - \alpha\beta)A(k^*)^\alpha$.

Then $\lim_t \beta^t u'(c_t)k_{t+1} \rightarrow 0$.

3.1 Dynamic Programming

Our approach up to now has been to look for a sequence of real numbers $\{k_{t+1}\}_{t=0}^{\infty}$ that generates an optimal consumption plan.

An alternative approach is called dynamic programming. Key to dynamic programming is to think of dynamic decisions as being made not once and for all but recursively: time period by time period. That is, the problem of

$$\max_{x_1, x_2} f(x_1, x_2),$$

is decomposed into two subproblems:

$$\max_{x_2} f(x_1, x_2), \text{ for fixed } x_1$$

and

$$\max_{x_1} f(x_1, x_2^*(x_1)).$$

Formally, a dynamic programming problem is a problem with the following structure: one can capture all relevant information for the decision maker in a state variable.

1. State space X . $x \in X$ is called a state variable.
2. $\Gamma(x_t)$ describe the feasibility set for x_{t+1} .
3. Per-period return $F(x_t, x_{t+1})$.

Value function of the sequential problem

$$\begin{aligned} V^*(x_0) = & \sup_{\{x_{t+1}\}_{t=0}^{\infty}} \sum_t \beta^t F(x_t, x_{t+1}) \\ s.t. & \quad x_{t+1} \in \Gamma(x_t) \\ & \quad x_0 \text{ is given.} \end{aligned}$$

REMARK 6 *General principles for choosing state variables:*

1. x_t must be relevant to the optimization problem at t .
2. Once the level of state variable is known, It does not matter how it is obtained in the past. (state variable is a sufficient statistics)
3. If $x_t = x_s$ for $t \neq s$, then the problem starting at t is the same as starting at s .

A second formulation (which appears in Recursive Macroeconomic Theory, Section 3.1):

1. State space X . $x \in X$ is called a state variable.
2. A control variable u , and a transition law $x_{t+1} = g(x_t, u_t)$.
3. Per-period return $F(x_t, u_t)$.

EXAMPLE 5 *Growth model*

EXAMPLE 6 *Continuous-time Growth model: typically in Formulation 2:*

$$\begin{aligned} \max \quad & \int_0^\infty e^{-rt} u(c_t) dt \\ \text{subject to} \quad & \frac{dk}{dt} = f(k_t) - \delta k_t - c_t \end{aligned}$$

EXAMPLE 7 *tree-cutting problem: Suppose a farm has a tree of size x_t . In each period t , the farmer can choose either to cut the tree and sell it for px_t , or wait one more period. If the farmer waits, the tree grows according to $x_{t+1} = f(x_t)$; otherwise $x_{t+1} = 0$. The state space is $X = R_+ \cup \{0\}$*

Formulation 1:

$$\begin{aligned} \Gamma(x) &= \{0, f(x)\}, \text{ if } x > 0 \\ \Gamma(x) &= \{0\}, \text{ if } x = 0 \\ F(x_t, x_{t+1}) &= 0 \text{ if } x_{t+1} = f(x_t) \\ F(x_t, x_{t+1}) &= px_t \text{ if } x_{t+1} = 0 \end{aligned}$$

3.1.1 Bellman equation, Functional equation, DPE.

Functional equation / Bellman equation / Dynamic Programming Equation

$$v(x_0) = \max_{x_1 \in \Gamma(x_0)} F(x_0, x_1) + \beta v(x_1), \forall x_0 \in X.$$

or

$$v(x_0) = \max_u r(x_0, u) + \beta v(g(x_0, u)), \forall x_0 \in X.$$

THEOREM 4 (THEOREM 4.2 IN SLP (STOKEY, LUCAS AND PRESCOTT)) *V^* satisfies the Functional Equation.*

THEOREM 5 (THEOREM 4.3 IN SLP) *If v satisfies the FE and for all x_0 and $(x_0, x_1, \dots, x_t, \dots) \in \Pi(x_0)$,*

$$\lim_{t \rightarrow \infty} \beta^t v(x_t) = 0,$$

then $v = V^$.*

EXAMPLE 8 *Find out $V^*(k_0)$ below and verify theorem 4.*

$$\begin{aligned} V^*(k_0) &= \sup_{\{k_{t+1}\}_{t=0}^{\infty}} \sum_t \beta^t \log(Ak_t^\alpha - k_{t+1}) \\ \text{s.t.} \quad &k_{t+1} \in [0, Ak_t^\alpha] \\ &k_0 \text{ is given.} \end{aligned}$$

Recall the policy $k_{t+1} = \alpha\beta Ak_t^\alpha$, or $\log(k_t) - \log(k^) = \alpha^t(\log(k_0) - \log(k^*))$. Compute*

the value function

$$\begin{aligned}
V^*(k_0) &= \sum_t \beta^t \log(Ak_t^\alpha - k_{t+1}) \\
&= \sum_t \beta^t \log((1 - \alpha\beta)Ak_t^\alpha) \\
&= \sum_t \beta^t \log((1 - \alpha\beta)A) + \sum_t \beta^t \alpha \log(k_t) \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \sum_t \beta^t \alpha [\log(k^*) + \alpha^t (\log(k_0) - \log(k^*))] \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha \log(k^*)}{1 - \beta} + \sum_t \beta^t \alpha \alpha^t (\log(k_0) - \log(k^*)) \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha \log(k^*)}{1 - \beta} + \frac{\alpha (\log(k_0) - \log(k^*))}{1 - \alpha\beta} \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha\beta \log(A\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\alpha \log(k_0)}{1 - \alpha\beta}
\end{aligned}$$

Then verify the Bellman equation.

$$\begin{aligned}
\text{right - side} &= \max_{k'} \log(Ak^\alpha - k') + \beta \left[\frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha\beta \log(A\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\alpha \log(k')}{1 - \alpha\beta} \right] \\
&= \log((1 - \alpha\beta)Ak^\alpha) + \beta \left[\frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha\beta \log(A\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\alpha \log(\alpha\beta Ak^\alpha)}{1 - \alpha\beta} \right] \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \alpha \log(k) + \beta \left[\frac{\alpha\beta \log(A\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\alpha \log(\alpha\beta A)}{1 - \alpha\beta} + \frac{\alpha^2 \log(k)}{1 - \alpha\beta} \right] \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha\beta \log(A\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \alpha \log(k) + \beta \left[\frac{\alpha^2 \log(k)}{1 - \alpha\beta} \right] \\
&= \frac{\log((1 - \alpha\beta)A)}{1 - \beta} + \frac{\alpha\beta \log(A\alpha\beta)}{(1 - \beta)(1 - \alpha\beta)} + \frac{\alpha \log(k)}{1 - \alpha\beta} \\
&= \text{left - side}
\end{aligned}$$

EXAMPLE 9 $X = R_+, F \equiv 0, \Gamma(x) = [x, x + 1]$. $V^* \equiv 0$. But the FE allows for other solutions besides V^* .

Further exploration of Bellman Equation

1. The optimal policy depends only on x .

$$V^*(x_0) = \max_{x_1 \in \Gamma(x_0)} F(x_0, x_1) + \beta V^*(x_1), \forall x_0 \in X.$$

$$V^*(x_0) = \max_u r(x_0, u) + \beta V^*(g(x_0, u)), \forall x_0 \in X.$$

The collection of maximizers in BE, $h(x_0)$, is called the policy correspondence. When the maximizer is unique, it is called policy function.

2. FOC conditions for interior solutions

$$0 = \frac{\partial F(x_0, x_1)}{\partial x_1} + \beta (V^*)'(x_1), \forall x_0 \in X.$$

Hence

$$V'(x_0) = \frac{\partial F(x_0, x_1)}{\partial x_0},$$

which is called Envelope theorem.

Combining the Envelope Theorem with the first-order condition, we reproduce the Euler equation in the growth model.

3. value function iteration (Section 3.1.1 and 3.1.2 in Sargent and Ljungqvist)

DEFINITION 1 contraction mapping operator $T : B(X) \rightarrow B(X)$.

$$(Tf)(x) = \max_{y \in \Gamma(x)} F(x, y) + \beta f(y)$$

THEOREM 6 (VALUE FUNCTION ITERATION) *If the per-period return function is bounded, then T maps from a bounded function to a bounded function. $T^j(v)$ converges to a fixed point of T uniformly for any bounded function v . This fixed-point is bounded and hence it is V^* .*

EXAMPLE 10 $X = R_+$, $F \equiv 0$, $\Gamma(x) = [x, x + 1]$. $V^* \equiv 0$.

EXAMPLE 11 *Consider the growth model with cobb-douglas production function and log utility. If the initial guess is $v(k) \equiv 0$, then we can compute the sequence $T^n v(k)$. If $v(k) = E_j + F_j \log(k)$, then $k_{t+1} = \frac{\beta F_j}{1 + \beta F_j} A k_t^\alpha$, and*

$$Tv(k) = E_{j+1} + F_{j+1} \log(k),$$

where

$$\begin{aligned} E_{j+1} &= \log\left(\frac{A}{1 + \beta F_j}\right) + \beta F_j \log\left(\frac{A \beta F_j}{1 + \beta F_j}\right) + \beta E_j \\ F_{j+1} &= \alpha + \alpha \beta F_j. \end{aligned}$$

First we solve for F_j .

$$\begin{aligned} F_0 &= 0 \\ \beta F_1 &= \alpha \beta \\ \beta F_2 &= \alpha \beta + (\alpha \beta)^2 \\ \beta F_3 &= \alpha \beta + (\alpha \beta)^2 + (\alpha \beta)^3 \end{aligned}$$

$$\begin{aligned}
\beta F_4 &= \alpha\beta + (\alpha\beta)^2 + (\alpha\beta)^3 + (\alpha\beta)^4 \\
&\dots \\
\beta F_\infty &= \frac{\alpha\beta}{1 - \alpha\beta}.
\end{aligned}$$

Second we solve for E_j .

$$\begin{aligned}
E_1 &= \log(A) \\
E_2 &= \log\left(\frac{A}{1 + \beta F_1}\right) + \beta F_1 \log\left(\frac{A\beta F_1}{1 + \beta F_1}\right) + \beta \log(A) \\
E_3 &= \log\left(\frac{A}{1 + \beta F_2}\right) + \beta F_2 \log\left(\frac{A\beta F_2}{1 + \beta F_2}\right) + \\
&\quad \beta(\log\left(\frac{A}{1 + \beta F_1}\right) + \beta F_1 \log\left(\frac{A\beta F_1}{1 + \beta F_1}\right) + \beta \log(A)) \\
&= \log\left(\frac{A}{1 + \beta F_2}\right) + \beta \log\left(\frac{A}{1 + \beta F_1}\right) + \beta^2 \log(A) + \\
&\quad \beta F_2 \log\left(\frac{A\beta F_2}{1 + \beta F_2}\right) + \beta(\beta F_1 \log\left(\frac{A\beta F_1}{1 + \beta F_1}\right)) \\
E_4 &= \dots \\
&\dots \\
E_{100} &= \log\left(\frac{A}{1 + \beta F_{99}}\right) + \beta \log\left(\frac{A}{1 + \beta F_{98}}\right) + \beta^2 \log\left(\frac{A}{1 + \beta F_{97}}\right) + \dots \\
&\quad + \beta F_{99} \log\left(\frac{A\beta F_{99}}{1 + \beta F_{99}}\right) + \beta(\beta F_{98} \log\left(\frac{A\beta F_{98}}{1 + \beta F_{98}}\right)) + \dots \\
&\dots
\end{aligned}$$

In Assignment 3, you are asked to show that

$$\lim_{j \rightarrow \infty} E_j = \frac{\log\left(\frac{A}{1 + \beta F_\infty}\right)}{1 - \beta} + \frac{\beta F_\infty \log\left(\frac{A\beta F_\infty}{1 + \beta F_\infty}\right)}{1 - \beta}.$$

DEFINITION 2 A function is strictly concave if for any x_1, x_2 in the domain, $x_1 \neq x_2$, and $\lambda \in (0, 1)$,

$$\lambda f(x_1) + (1 - \lambda)f(x_2) < f(\lambda x_1 + (1 - \lambda)x_2)$$

THEOREM 7 (STRICT CONCAVITY) *If production function and utility function are strictly concave and strictly increasing, so is the value function.*

PROOF OF :

- (a) If f is weakly concave, I will show that Tf is strictly concave. Pick $k^1 < k^2$, $\lambda \in (0, 1)$,

$$\begin{aligned} Tf(k^1) &= \max_{k'} u(F(k^1, 1) + (1 - \delta)k^1 - k') + \beta f(k') \\ &= u(F(k^1, 1) + (1 - \delta)k^1 - k^{1*}) + \beta f(k^{1*}) \\ Tf(k^2) &= \max_{k'} u(F(k^2, 1) + (1 - \delta)k^2 - k') + \beta f(k') \\ &= u(F(k^2, 1) + (1 - \delta)k^2 - k^{2*}) + \beta f(k^{2*}) \end{aligned}$$

$$\begin{aligned} Tf(\lambda k^1 + (1 - \lambda)k^2) &= \max_{k'} u(F(\lambda k^1 + (1 - \lambda)k^2, 1) + (1 - \delta)(\lambda k^1 + (1 - \lambda)k^2) - k') \\ &\quad + \beta f(k') \\ &\geq u(F(\lambda k^1 + (1 - \lambda)k^2, 1) + (1 - \delta)(\lambda k^1 + (1 - \lambda)k^2) \\ &\quad - (\lambda k^{1*} + (1 - \lambda)k^{2*})) + \beta f(\lambda k^{1*} + (1 - \lambda)k^{2*}) \\ &> u(\lambda F(k^1, 1) + (1 - \lambda)F(k^2, 1) + (1 - \delta)(\lambda k^1 + (1 - \lambda)k^2) \\ &\quad - (\lambda k^{1*} + (1 - \lambda)k^{2*})) + \beta f(\lambda k^{1*} + (1 - \lambda)k^{2*}) \\ &\geq \lambda u(F(k^1, 1) + (1 - \delta)k^1 - k^{1*}) + \\ &\quad (1 - \lambda)u(F(k^2, 1) + (1 - \delta)k^2 - k^{2*}) + \beta f(\lambda k^{1*} + (1 - \lambda)k^{2*}) \\ &\geq \lambda Tf(k^1) + (1 - \lambda)Tf(k^2) \end{aligned}$$

- (b) $\lim_n T^n(f)$ uniformly converges to V^* . Hence V^* is weakly concave.

- (c) But $V^* = TV^*$, hence V^* is strictly concave.

Q.E.D.